

UNIT 4: Boolean Algebra

1. Role of Logical Operations in Computing

Logical operations are used in computing to:

- Make decisions
- Perform comparisons
- Design digital circuits
- Control program flow

Computers work on **binary values (0 and 1)**.

2. Basic Terms

Binary-Valued Quantities

Quantities that have only two values:

- 0 (False)
- 1 (True)

Boolean Variable

A variable that can store only 0 or 1.

Boolean Constant

Fixed values:

- 0
 - 1
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3. Boolean Operators

Operator	Symbol	Meaning
AND	.	Multiplication
OR	+	Addition
NOT	'	Complement

4. Truth Tables

AND Operation

A	B	A.B
0	0	0
0	1	0
1	0	0
1	1	1

OR Operation

A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

NOT Operation

A	A'
0	1
1	0

5. Laws of Boolean Algebra

- Closure Law
 - Commutative Law
 - Associative Law
 - Identity Law
 - Inverse Law
 - Principle of Duality
 - Idempotent Law
 - Distributive Law
 - Absorption Law
 - Involution Law
 - DeMorgan's Law
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Important Laws

Commutative Law

- $A + B = B + A$
- $A.B = B.A$

Identity Law

- $A + 0 = A$
- $A.1 = A$

Inverse Law

- $A + A' = 1$
- $A.A' = 0$

Idempotent Law

- $A + A = A$
- $A.A = A$

DeMorgan's Law

- $(A.B)' = A' + B'$
 - $(A + B)' = A'.B'$
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6. SOP and POS Forms

Sum of Products (SOP)

Expression written as sum (OR) of product (AND) terms.

Product of Sums (POS)

Expression written as product (AND) of sum (OR) terms.

7. Minimization of Boolean Expressions

- Reduce expression using Boolean laws
 - Convert to minimal SOP or POS form
 - Use Karnaugh Map (K-Map)
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8. Karnaugh Map (K-Map)

Used to simplify Boolean expressions.

- Works up to 4 variables
 - Group 1's in pairs, quads, octets
 - Write simplified expression
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9. Application of Boolean Logic

(A) Digital Electronic Circuits

Basic Logic Gates:

- NOT Gate
- AND Gate
- OR Gate
- NAND Gate
- NOR Gate

(B) Boolean Operators in SQL

Used in SELECT statements:

- AND
- OR
- NOT

(C) Search Engine Queries

Example:

- Computer AND Science
- Computer OR Mathematics

Very Important Exam Questions

- Write truth tables of AND, OR, NOT
- State and prove DeMorgan's Theorem
- Simplify Boolean expression
- Define SOP and POS
- Explain K-Map with example

Exam Tips

- Draw truth tables properly
- Write laws clearly with formula
- Show steps while simplifying expressions
- Circle groups correctly in K-Map